

29.4 All subsets and stepwise search

With m candidate regressors and the intercept always in, there are 2^m subsets. This section describes exhaustive search, branch and bound and the stepwise shortcuts, and shows that the subset they find depends on small features of the data, far more than the fitted values do.

29.4.1 Searching all subsets

The search splits into two stages, because every criterion based on the residual sum of squares agrees within a size.

Proposition 29.4.1. One size, one ranking

Among candidates with the same number of columns p , the orderings by C_p , AIC, AICc, BIC, R^2 , $\bar{R}^2$ and SSE coincide.

Proof. For fixed p , n and s^2 , each of C_p , $n \log(\text{SSE}/n)$ plus a penalty depending on p , $1 - \text{SSE}/\text{SST}$ and $1 - (n-1)\text{SSE}/\{(n-p)\text{SST}\}$ is a monotone function of SSE, increasing for the criteria to be minimized and decreasing for the two R^2 .

■

So it suffices to find, for each size, the subset with the smallest residual sum of squares, and then compare the $m+1$ winners with the criterion. Cross-validation does not reduce in this way, since its value is not a function of SSE and p ; in practice it is applied to the size winners, as in Example (Cross-validating the state regressions).

Visiting all 2^m subsets is cheap per subset. Sweeping on one pivot (Definition 10.2.1) adds or removes a column in $O(m^2)$ operations, and a Gray-code order, in which consecutive subsets differ in one column, gives every residual sum of squares with one sweep (Schatzoff, Tsao and Fienberg 1968). That makes m up to about 20 routine. Beyond that the number of subsets is the obstacle: finding the best subset of a given size is NP-hard in general (Natarajan 1995).

29.4.2 Branch and bound

The key fact is that removing columns never reduces the residual sum of squares: if $\mathcal{C}(\mathbf{X}_A) \subseteq \mathcal{C}(\mathbf{X}_B)$, then $\text{SSE}_A - \text{SSE}_B = \|(\mathbf{M}_B - \mathbf{M}_A)\mathbf{y}\|^2 \geq 0$ by Theorem 6.5.1. A subset whose residual sum of squares is already too large therefore cannot have a good subset below it.

Organize the subsets as a tree. The root is the full set $\{1, \dots, m\}$. A node is a subset S together with the index ℓ of the last column deleted to reach it ($\ell = 0$ at the root), and its children delete one more column $j \in S$ with $j > \ell$. Deleting in increasing order of index reaches every subset along exactly one path. The descendants of (S, ℓ) are the subsets $T \subseteq S$ that contain $F = \{j \in S : j \leq \ell\}$, the columns that can no longer be deleted.

Proposition 29.4.2. Branch and bound for C_p

Let c^* be the smallest C_p found so far. If a node (S, ℓ) satisfies

$$\frac{\text{SSE}_S}{s^2} - n + 2(|F| + 1) \geq c^*,$$

then no descendant of the node has $C_p < c^*$, and the node need not be expanded. A depth-first search that evaluates every expanded node and skips such nodes' descendants returns a subset of minimal C_p .

Proof. A descendant T contains F and is contained in S , so $\text{SSE}_T \geq \text{SSE}_S$ and $|T| \geq |F|$, with the intercept counted separately. Hence its $C_p = \text{SSE}_T/s^2 - n + 2(|T| + 1)$ is at least the left side of the display. The search only skips subsets that cannot beat a value already attained, so the minimum it reports is the minimum over all subsets. ■

The same bound works for AIC, BIC or the best subset of each size. Furnival and Wilson (1974) combined bounds of this kind with an efficient order of sweeps in the “leaps and bounds” algorithm, which handles 30 to 40 regressors. Bertsimas, King and Mazumder (2016) solved much larger problems by mixed-integer optimization, with a certificate of optimality.

Example. How much branch and bound saves

Take $n = 100$ cases and $m = 15$ independent normal regressors, five with nonzero slopes (1, -0.8, 0.6, 0.4, 0.3) and ten with zero slopes. Exhaustive search evaluates 32768 subsets. The branch and bound of [Proposition 29.4.2](#) finds the same C_p -best subset, with 6 regressors (the four largest slopes and two irrelevant columns), after 1110 evaluations, a fraction 0.034 of the work.

```
import itertools
import numpy as np
rng = np.random.default_rng(2908)
n, m = 100, 15
X = rng.normal(size=(n, m))
beta = np.r_[1.0, -0.8, 0.6, 0.4, 0.3, np.zeros(m - 5)]
y = 2 + X @ beta + rng.normal(size=n)
Xc, yc = X - X.mean(axis=0), y - y.mean()           # centring accounts for
the intercept
G, g = Xc.T @ Xc, Xc.T @ yc

def sse(S):
    S = list(S)
    if not S:
        return yc @ yc
    return yc @ yc - g[S] @ np.linalg.solve(G[np.ix_(S, S)], g[S])

s2 = sse(range(m)) / (n - m - 1)

def cp(S):
    return sse(S) / s2 - n + 2 * (len(S) + 1)
```

```

def branch_and_bound():
    """Children delete a column with a larger index than the last one deleted, so
    every
    subset is visited at most once; a node is not expanded when no descendant can
    win."""
    best = [np.inf, None]
    count = [0]

    def visit(S, last):
        e = sse(S)
        count[0] += 1
        value = e / s2 - n + 2 * (len(S) + 1)
        if value < best[0]:
            best[:] = [value, S]
            forced = [j for j in S if j <= last] # kept by every descendant
            if e / s2 - n + 2 * (len(forced) + 1) >= best[0]:
                return # SSE only grows as
            # columns go
            for j in S:
                if j > last:
                    visit(tuple(k for k in S if k != j), j)

    visit(tuple(range(m)), -1)
    return best[1], best[0], count[0]

S_bb, cp_bb, fits = branch_and_bound()
print("branch and bound:", S_bb, round(cp_bb, 3), "after", fits, "fits of", 2**m)

```

[Run this code online](#)

29.4.3 Stepwise search

Stepwise methods give up the guarantee of finding the best subset in exchange for examining only $O(m^2)$ of them.

Definition 29.4.1. Stepwise selection

Let the current subset be S , with p_S columns including the intercept.

- (a) **Forward selection** starts from the intercept alone. At each step it adds the column $j \notin S$ that most reduces SSE, provided its **F-to-enter** $F_j = (\text{SSE}_S - \text{SSE}_{S+j}) / \{\text{SSE}_{S+j} / (n - p_S - 1)\}$ exceeds a threshold F_{in} , and stops otherwise.
- (b) **Backward elimination** starts from the full model. At each step it removes the column $j \in S$ with the smallest **F-to-remove** $F_j = (\text{SSE}_{S-j} - \text{SSE}_S) / \{\text{SSE}_S / (n - p_S)\}$, provided this is below a threshold F_{out} , and stops otherwise.
- (c) **Stepwise regression** (Efroymson 1960) alternates: after each forward step it removes any column whose **F-to-remove** has fallen below F_{out} , with $F_{\text{out}} \leq F_{\text{in}}$.

The thresholds may be replaced by a criterion: add or remove the column that lowers AIC (or C_p , or BIC) most, and stop when no move lowers it.

The single step of forward selection can be described in several equivalent ways, all of them versions of the Frisch–Waugh–Lovell theorem.

Proposition 29.4.3. What a stepwise step maximizes

Let $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_S)\mathbf{y}$ and $\tilde{\mathbf{x}}_j = (\mathbf{I} - \mathbf{M}_S)\mathbf{x}_j \neq \mathbf{0}$, and let $r_j = \tilde{\mathbf{y}}^\top \tilde{\mathbf{x}}_j / (\|\tilde{\mathbf{y}}\| \|\tilde{\mathbf{x}}_j\|)$ be the partial correlation of $\mathbf{y}$ and $\mathbf{x}_j$ given the columns of S (Definition 14.5.1).

- (a) $\text{SSE}_S - \text{SSE}_{S+j} = r_j^2 \text{SSE}_S$ and $F_j = (n - p_S - 1) r_j^2 / (1 - r_j^2)$. So the column that most reduces SSE is the one with the largest r_j^2 , the largest F -to-enter, and the smallest C_p , AIC or BIC of the enlarged model.
- (b) For $j \in S$, $\text{SSE}_{S-j} - \text{SSE}_S = t_j^2 s_S^2$, where t_j is the t statistic of $\mathbf{x}_j$ in the model S and $s_S^2 = \text{SSE}_S / (n - p_S)$. So backward elimination removes the column with the smallest $|t_j|$, and its F -to-remove is t_j^2 .

Proof.

- (a) By Theorem 6.6.2(b), the residual vector of the model $S + j$ is the residual of the regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{x}}_j$, so $\text{SSE}_{S+j} = \|\tilde{\mathbf{y}}\|^2 - (\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}_j\|^2 = (1 - r_j^2) \text{SSE}_S$. Then $F_j = (n - p_S - 1) r_j^2 / (1 - r_j^2)$ is increasing in r_j^2 . At a fixed size the criteria rank by SSE (Proposition 29.4.1).
- (b) Apply (a) to the model $S - j$ and the column $\mathbf{x}_j$: the increase is $(\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}_j\|^2$ with residuals now taken on $S - j$. By Theorem 6.6.2(a) this is $\hat{\beta}_j^2 \|\tilde{\mathbf{x}}_j\|^2$, and by Eq. 6.6.2 $\text{Var}(\hat{\beta}_j) = \sigma^2 / \|\tilde{\mathbf{x}}_j\|^2$, so the increase is $\hat{\beta}_j^2 s_S^2 / \widehat{\text{Var}}(\hat{\beta}_j) = t_j^2 s_S^2$. ■

Criterion-based stopping is therefore threshold-based stopping with a particular threshold: forward selection by AIC adds a column iff its F -to-enter exceeds about 2, and by BIC iff it exceeds about $\log n$ (the table in Section 29.2). The traditional thresholds $F_{\text{in}} = F_{\text{out}} = 4$ correspond to BIC at $n \approx 55$.

For the state data, forward selection by AIC adds single parenthood and then urbanization and stops. Backward elimination by AIC removes white, high school, poverty (in that order) and stops at the same two regressors, which are also the all-subsets AIC choice. Forward and backward computed the criterion for 13 and 15 subsets, against 32 for exhaustive search. Agreement is common but not guaranteed.

Example. A pair that forward selection cannot find

Take $n = 60$ cases of three regressors, $\mathbf{x}_1$ and $\mathbf{x}_2$ with sample correlation 0.96 and $\mathbf{x}_3$ independent of both, and let $\mathbf{y} = 3(\mathbf{x}_1 - \mathbf{x}_2) + 0.4\mathbf{x}_3 + \boldsymbol{\varepsilon}$. The signal lies mostly in the difference $\mathbf{x}_1 - \mathbf{x}_2$, a direction in which the regressors vary little. The sample correlations of $\mathbf{y}$ with $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ are -0.02 , -0.19 and 0.41 . Forward selection enters $\mathbf{x}_3$ first and then $\mathbf{x}_2$, reaching $R^2 = 0.19$. The best pair is $\{\mathbf{x}_1, \mathbf{x}_2\}$, with $R^2 = 0.39$. Neither of its members looks useful alone, so a procedure that adds one column at a time never sees it. Backward elimination starts from a model that contains the pair and can keep it, but it has blind spots of its own (Exercise (C1)) and cannot start when $m \geq n$.

```
import itertools
```

```

import numpy as np
rng = np.random.default_rng(2909)
n2 = 60
z = rng.normal(size=(n2, 3))
x1 = z[:, 0]
x2 = 0.95 * z[:, 0] + np.sqrt(1 - 0.95**2) * z[:, 1]      # corr(x1, x2) about 0.95
x3 = z[:, 2]
Xf = np.column_stack([x1, x2, x3])
yf = 3 * (x1 - x2) + 0.4 * x3 + rng.normal(size=n2)      # the signal is in x1 - x2

def rss(cols):
    A = np.column_stack([np.ones(n2)] + [Xf[:, j] for j in cols])
    b, *_ = np.linalg.lstsq(A, yf, rcond=None)
    return np.sum((yf - A @ b) ** 2)

first = min(range(3), key=lambda j: rss([j]))
second = min((j for j in range(3) if j != first), key=lambda j: rss([first, j]))
best_pair = min(itertools.combinations(range(3), 2), key=rss)
print("forward chooses", sorted([first, second]), "RSS", round(rss([first, second]),
1))
print("best pair", list(best_pair), "RSS", round(rss(best_pair), 1))

```

[Run this code online](#)

Collinear regressors are exactly where this happens: a contrast between them can be highly predictive while each member is not (Chapter 26 describes such directions through the eigenvectors of $\mathbf{X}^\top \mathbf{X}$).

29.4.4 Comparing the procedures

Example. Selection procedures in a simulation

Take $n = 60$ cases of eight normal regressors with all pairwise correlations 0.5, $\sigma = 1$, an intercept, and three nonzero slopes, either weak (1, 0.5, 0.25) or strong (1, 0.8, 0.6). Each procedure is scored by its loss at the design points, $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 / \sigma^2$, with the intercept's expected contribution 1 included. Over 2000 data sets:

Weak slopes:

procedure	loss	Pr(true)	size
least squares, all eight	8.98	0.00	8.00
true subset (oracle)	3.97	1.00	3.00
all subsets, AIC	7.81	0.18	3.42
all subsets, BIC	7.33	0.18	2.52
all subsets, leave-one-out	7.76	0.19	3.34
forward, AIC	7.70	0.19	3.34
backward, AIC	7.89	0.17	3.50

Strong slopes:

procedure	loss	Pr(true)	size
least squares, all eight	9.02	0.00	8.00
true subset (oracle)	4.06	1.00	3.00
all subsets, AIC	7.27	0.37	3.93
all subsets, BIC	6.17	0.70	3.20
all subsets, leave-one-out	7.17	0.39	3.85
forward, AIC	7.16	0.39	3.86
backward, AIC	7.35	0.35	3.98

Every selection rule beats least squares on all eight regressors, whose expected loss is 9 (Theorem 29.1.1(c)), but none comes close to the oracle's 4. The search method hardly matters: forward selection by AIC returned the all-subsets AIC choice in a fraction 0.95 of the weak data sets and 0.96 of the strong ones. The criterion matters more: with strong effects BIC finds the true subset in a fraction 0.70 of data sets and has the smallest loss, while with a weak third slope every rule usually misses it, BIC keeping 2.52 regressors on average.

The simulation illustrates Proposition 29.2.4 in a finite sample: where a small true model exists and its effects are large, BIC wins; with many small effects the comparison tilts towards AIC and cross-validation.

29.4.5 Instability

A procedure is **unstable** if small changes in the data change its output a great deal. Subset selection is a discontinuous function of the data: a column is in or out. Breiman (1996b) showed that this instability inflates the variance of the predictions that follow, and that averaging over perturbed data sets can reduce it.

Example. Bootstrapping the state regressions

Resample the 50 states with replacement (the case bootstrap of [Section 23.1](#)) 1000 times and repeat the all-subsets AIC and BIC choices on each resample. AIC chooses 27 different subsets of the 32 and BIC 25. The AIC choice for the original data, single parenthood and urbanization, is chosen in a fraction 0.15 of resamples, and the most frequent AIC choice, poverty with single parenthood and urbanization, in 0.20. The BIC choice, single parenthood alone, is its most frequent choice, at 0.31. [Figure 29.4.1](#) shows how often each regressor is included. Even single parenthood, whose t statistic in the chosen model is about 5.5, is left out by AIC in a fifth of the resamples. It is correlated -0.70 with the percentage of white residents, and the two can stand in for each other.

```
import itertools
from collections import Counter
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
names = ["hs_grad", "poverty", "single", "white", "urban"]
y_all = np.log(data["violent"].to_numpy())
Z_all = data[names].to_numpy()
```

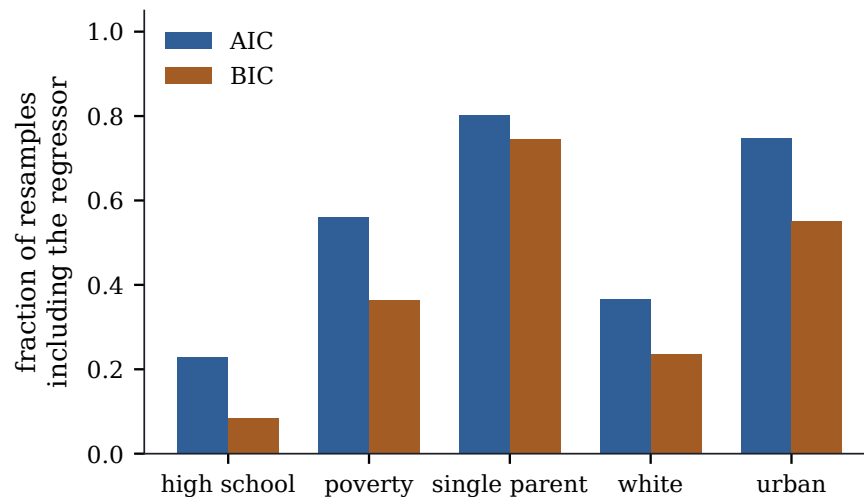


Figure 29.4.1: Fraction of 1000 bootstrap resamples of the 50 states in which each regressor is in the subset chosen by AIC or BIC (log violent crime, all 32 subsets).

```

n = len(y_all)
subsets = [S for k in range(6) for S in itertools.combinations(range(5), k)]

def select(Z, y, penalty):
    def crit(S):
        X = np.column_stack([np.ones(len(y)), Z[:, list(S)]])
        b, *_ = np.linalg.lstsq(X, y, rcond=None)
        return n * np.log(np.sum((y - X @ b) ** 2) / n) + penalty * (len(S) + 2)
    return min(subsets, key=crit)

rng = np.random.default_rng(2911)
B = 200                                     # the book's figure
    uses 1000
chosen = {"AIC": Counter(), "BIC": Counter()}
for _ in range(B):
    idx = rng.integers(0, n, size=n)         # resample whole
    states
    chosen["AIC"][select(Z_all[idx], y_all[idx], 2.0)] += 1
    chosen["BIC"][select(Z_all[idx], y_all[idx], np.log(n))] += 1
for crit, counts in chosen.items():
    print(crit, "distinct models:", len(counts))
    for S, c in counts.most_common(3):
        print("    ", [names[j] for j in S], c / B)

```

[Run this code online](#)

Instability is a property of the question as much as of the method. The three models favoured by the criteria and the bootstrap (single parenthood alone, with urbanization, and with poverty and urbanization) predict almost equally well (their leave-one-out errors in Example (Cross-validating the state regressions) lie between 0.1129 and 0.1184), so any rule that must name one will name different ones for similar data. The remedies are to report the uncertainty, as in the figure, or to stop requiring a single model: average over bootstrap resamples (bagging, Breiman 1996a) or over models weighted by posterior probability (Hoeting, Madigan, Raftery and Volinsky 1999), or shrink instead of selecting (Chapter 27).

29.4.6 Exercises

29.4.6.1 A. Check your understanding

Exercise. A1

How many subsets must exhaustive search examine for $m = 20$ and $m = 40$ candidate regressors? At a million subsets per second, how long does each take?

29.4.6.2 B. Practice

Exercise. B1

Show that in the tree of [Proposition 29.4.2](#) every subset of $\{1, \dots, m\}$ is reached along exactly one path from the root, and that the descendants of (S, ℓ) are exactly the subsets T with $F \subseteq T \subseteq S$.

Solution. A subset T with complement $\{j_1 < \dots < j_r\}$ in $\{1, \dots, m\}$ is reached by deleting $j_1, \dots, j_r$ in increasing order, and every path deletes in increasing order, so this is the only path. A descendant of (S, ℓ) deletes further columns of S with indices above ℓ , so it contains $F = \{j \in S : j \leq \ell\}$; conversely any T with $F \subseteq T \subseteq S$ is reached from (S, ℓ) by deleting the columns of $S \setminus T$, all of which exceed ℓ , in increasing order.

Exercise. B2

In stepwise regression, show that the column just entered has F -to-remove equal to its F -to-enter. Deduce that it cannot be removed at the next step if $F_{\text{out}} \leq F_{\text{in}}$, and explain what can happen if $F_{\text{out}} > F_{\text{in}}$.

Solution. After j enters, the model is $S + j$ with $p_S + 1$ columns, and its F -to-remove is $(\text{SSE}_S - \text{SSE}_{S+j}) / \{\text{SSE}_{S+j} / (n - p_S - 1)\}$, which is the F -to-enter of j at the previous step. It exceeded $F_{\text{in}} \geq F_{\text{out}}$, so j stays. If $F_{\text{out}} > F_{\text{in}}$, a column with $F_{\text{in}} < F_j < F_{\text{out}}$ is added and immediately removed, returning to S , where it is added again: the procedure cycles.

Exercise. B3

Show that a column's F -to-remove in backward elimination is the square of its t statistic, and that backward elimination with $F_{\text{out}} = t_{\alpha/2}^2(\nu)$ is the rule “drop the least significant coefficient while any is not significant at level α ”. Why do the p -values of the surviving coefficients not have their nominal meaning?

29.4.6.3 C. Going deeper

Exercise. C1

In [Example \(A pair that forward selection cannot find\)](#), show that the best subset of size one is $\{\mathbf{x}_3\}$ and the best of size two is $\{\mathbf{x}_1, \mathbf{x}_2\}$, so the best subsets of consecutive sizes need

not be nested. Why does this not contradict Theorem 6.5.1?

Solution. Among single columns the residual sum of squares is smallest for the largest squared correlation with $\mathbf{y}$, and 0.41^2 is far above 0.19^2 and 0.02^2 . The best pair was computed in the example. Theorem 6.5.1 compares a subset with its own supersets, so $\text{SSE}_{\{3\}} \geq \text{SSE}_{\{1,3\}}$ and $\text{SSE}_{\{3\}} \geq \text{SSE}_{\{2,3\}}$; it says nothing about $\{1, 2\}$, which does not contain $\{3\}$. Because best subsets need not be nested, a search that only adds columns (forward) or only removes them (backward) can miss the best subset of some size.